

JULIUS GRAF

📍 Berkeley, CA | 📞 +1 (510) 859-5718 | ✉️ julius.graf@berkeley.edu | 🌐 juliusgraf | in julius-graf | 🌐 Website

Academic Summary: Ph.D. student in Industrial Engineering and Operations Research at UC Berkeley focused on stochastic control, market microstructure, and machine learning for financial markets. Research includes market making, order-flow modeling, filtering under partial information, and reinforcement learning.

EDUCATION

University of California, Berkeley

PhD in Industrial Engineering and Operations Research (Expected May 2030)

California, USA

07/2025 - Present

- **Research:** Stochastic control, cyber risk, financial microstructure, filtering, Hawkes processes, statistical learning for order-flow modeling with Prof. Thibaut Mastrolia
- **Coursework:** Fractional Brownian Motion and Malliavin Calculus, Signature Transform

University of California, Berkeley

MEng in Industrial Engineering and Operations Research, FinTech Concentration – GPA 3.92/4.0

California, USA

08/2024 - 05/2025

- **Coursework:** Applications of Machine Learning to Electronic Markets, Theoretical Statistics, Financial Engineering, Applied Stochastic Processes, Network Flows and Graphs, Mathematical Programming, Stochastic Optimization for Machine Learning
- **Award:** UC Berkeley College of Engineering MEng Fung Excellence Scholarship (\$30k)

CentraleSupélec, Université Paris-Saclay

MSc& Eng in Applied Mathematics – GPA 3.97/4.0 (Top 1%)

Gif-sur-Yvette, France

09/2022 - 05/2025

- **Coursework:** Stochastic Finance, Markov Chains and Martingales, PDE, Probability, Quantum Physics

Lycee Louis-le-Grand

Preparatory Classes MPSI-MP*, Major: Mathematics, Minor: Physics — GPA 4.0/4.0

Paris, France

09/2020 - 08/2022

- **Award:** French Government's Major-Excellence Scholarship granted for academic results - 1 of 150 from 17,000 students worldwide

RESEARCH EXPERIENCE

Learning Market Making with Closing Auctions

UC Berkeley IEOR Department, Supervised by Prof. Thibaut Mastrolia

Berkeley, USA

06/2025 - Present

- **Publication:** "Learning Market Making with Closing Auctions" with T. Mastrolia, arXiv preprint, 2026
- **Presentation:** Presenting "Learning Market Making with Closing Auctions" at the Bachelier World Congress, Bologna, Summer 2026
- Developed a Deep Q-Learning framework for optimal market making with closing-auction liquidity, inventory risk, and anticipated clearing-price dynamics
- Built a generative stochastic market simulator in Python to evaluate trading policies across synthetic rough Heston paths and historical S&P 500 assets

Deep Reinforcement Learning for Automated Stock Trading

UC Berkeley IEOR Department, MEng Capstone Project Supervised by Lizeng Zhang

Berkeley, USA

10/2024 - 05/2025

- Built an ensemble reinforcement-learning framework combining PPO, A2C, and DDPG for systematic equity trading, trained on 10 years of DJIA data and evaluated using walk-forward out-of-sample testing.
- Designed capital-deployment and turnover regularization to control trading intensity and transaction-cost exposure, reducing turnover by 9% with limited degradation in out-of-sample Sharpe.

Lipschitz Regularization for Robust Neural Image Segmentation

Research Project Supervised by Prof. Jean-Christophe Pesquet, Center for Visual Computing

Gif-sur-Yvette, France

02/2023 - 01/2024

- Enhanced model robustness by reducing the Lipschitz constant from 10^8 to ~ 1 while maintaining 90% precision
- Implemented firm non-expansiveness CNN constraints in PyTorch and ran large-scale experiments on image-segmentation benchmarks

PROFESSIONAL EXPERIENCE

Founder, EquitX

AI-powered access layer for high-demand live events; Founders, Inc. Canopy Program

Berkeley, USA

01/2026 - Present

- Built MVP for predictive, budget-based matching that helps fans access scarce live-event inventory while giving organizers visibility into secondary demand and price sensitivity
- Ran first paid EquitX-powered live event during Founders Inc's Canopy Demo Day, selling 110 entries to real users and generating \$450 in revenue

Teaching Assistant, MICS, CentraleSupélec, Université Paris-Saclay

Research Laboratory in Mathematics and Computer Science, CentraleSupélec, Université Paris-Saclay

Gif-sur-Yvette, France

09/2023 - 06/2024

- Led weekly 4-hour tutorials in Analysis, Probability, and PDEs, explaining complex mathematical concepts to undergraduate students

SKILLS

- **Programming & Data:** Python, PyTorch, Scikit-Learn, NumPy, Pandas, SQL, R, MATLAB, L^AT_EX
- **Languages:** German (Native), French (Bilingual), English (Fluent), Russian (B1), Spanish (Beginner)